

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	29 September 2026
Team ID	Team No: 05
Project Name	AI WeatherWise
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration using name, email, and password. Validate required registration details. Create a new user account.
FR-2	User Login & Authentication	Login using registered email and password. Validate credentials. Generate JWT authentication token.
FR-3	Weather Information	Request weather information for a city. Retrieve current weather data through the weather API. Return weather data to the user.
FR-4	Favorite Locations	Save a user's favorite location. Associate the location with the user account. Retrieve saved locations when required.
FR-5	AI Weather Recommendation	Send weather information/context for AI processing. Generate weather-based insights and recommendations when AI service is available. Return a fallback recommendation when the external AI service is unavailable.
FR-6	API Request Processing & Error Handling	Route API requests through the Express backend. Apply authentication/validation middleware where required. Handle invalid requests and service errors through centralized error handling.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The REST API should provide clear endpoints and predictable request/response formats so that users and developers can use the weather services easily.
NFR-2	Security	User passwords are protected using bcryptjs. JWT is used for authentication. Sensitive configuration values are kept in environment variables.
NFR-3	Reliability	The backend should handle API and service errors gracefully. The AI recommendation route provides fallback behavior when the external AI service is unavailable.
NFR-4	Performance	The backend should process REST requests efficiently using a modular Node.js and Express.js service structure. No caching or CDN is specified in the current implementation.
NFR-5	Availability	The weather service depends on the availability of the external weather API. AI recommendations can fall back when the external AI service is unavailable.
NFR-6	Scalability	The backend separates routes, controllers, middleware, models, and services, allowing additional endpoints and services to be added as the project grows.